

The Water Cycle

The water cycle is the continuous movement of water within the Earth and atmosphere. It is a complex system that includes many different processes. Liquid water evaporates into water vapor, condenses to form clouds, and precipitates back to Earth in the form of rain or snow.

Evaporation occurs when the sun heats up water in rivers, lakes, and oceans and turns it into vapor or steam. The water vapor leaves the river, lake, or ocean and goes into the air. As shown in the diagram above, roughly 90% of atmospheric moisture comes from evaporation off the surface of oceans, with the remainder coming from transpiration in plants.

Condensation is the process by which water vapor in the air is changed into liquid water. Condensation is crucial to the water cycle because it is responsible for the formation of clouds. These clouds may produce precipitation, which is the primary route for water to return to the Earth's surface within the water cycle.

Precipitation occurs when so much water has condensed that the air cannot hold it anymore. The clouds become saturated with water droplets, and the droplets fall from the sky as rain, hail, sleet, or snow, depending on the temperature of the surrounding air.

Collection happens when water pools in large bodies like oceans, lakes, and rivers, e.g. the Amazon River or the Pacific Ocean. Some of the water is absorbed into the ground through a process called infiltration, where it becomes groundwater that plants and trees can use.

Groundwater eventually seeps back into rivers and streams, continuing the cycle. Scientists estimate that a single water molecule spends about nine days in the atmosphere before falling back to Earth, and can take thousands of years to move through an underground aquifer.

Human activity affects the water cycle in significant ways. Deforestation reduces transpiration, urbanization increases runoff by covering soil with impermeable surfaces, and climate change is altering evaporation rates and

precipitation patterns across the globe. Understanding the water cycle helps scientists predict weather, manage freshwater resources, and study the effects of climate change on ecosystems worldwide.